EFFECTIVE PROMPTING

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Start a new conversation for a new task

AI reads the entire conversation (the context window) each time you prompt. Irrelevant stuff consumes tokens and may be confusing.

When a conversation gets long

Tell AI to write a handoff document so you can start a fresh conversation. How long? There is a tradeoff; it depends on how much is irrelevant.

Managing conversations

Good enough prompting

Most ‘prompt engineering’ techniques from a few years ago are now either useless or actually counterproductive. Don’t tell it “you are an expert data scientist” or “think step by step.”

Tell it what you want, not what to do

I want … How can we do it?

Ask AI for help in prompting

If you are unsure how to ask something, ask AI how to ask it.

If something doesn’t work

Try again.

> Treat AI as an infinitely patient colleague with a poor memory.

Your go-to prompt

Think about the best way to do this, and let's develop a plan together.

Ex-post diagnosis

Prompt: Explain what you did step by step. Did you make any decisions that we have not discussed? Write this as an HTML doc.

Use AI to evaluate

Start a new conversation or ask for a subagent. Ask it to evaluate the first agent's work, summarized in the HTML doc, relative to your goals.

Plan, evaluate, critique

What this buys

1. AI is a prediction machine. Ask it to do something, and it will probably do what it has seen most often in its training data.
2. Ask it the best way to do something, and it will regurgitate evaluations of methods it has seen.
3. Ask it to proceed after explaining the best way, it will read the “the best way is …” in the conversation history and follow that.

4. Misunderstandings and unstated assumptions are the most common sources of errors. Discuss before and after to surface those.
5. AI is not good at critiquing itself in the same conversation, because it reads its reasoning in the conversation history and predicts the same thing. You need a fresh context window to get a reliable critique.

1. Ask AI to write a report on something (for example, effective prompting techniques). Have it save the report as an HTML file.
2. Start a new conversation (logout and log back in is the easiest way on the AI lab). Ask AI to read the file created in step #1 and to evaluate the report.

Try it

Beware of tunnel vision

Prediction also means momentum. Once Claude has started down a path, continuing it is the likeliest next step.

What you will see

- It hits a roadblock and tries to get around it
- Then another way around, and another — exhaustively, tenaciously
- What it rarely does is return to the start and ask whether a different path was the better one all along

What helps

- Periodically ask: are there other approaches that might work better for reaching the end goal?
- Restate the goal, not just the obstacle — that gives it permission to abandon the path instead of defending it

Don’t ask AI whether your idea is good. Ask it to make the strongest possible case that your idea is wrong.

Sycophancy

Training on human feedback teaches that agreement gets high ratings. Ask “is my plan sound?” and Claude finds reasons it is.

The fix

“What is the strongest argument against this acquisition?”

One step further

Have it play a skeptical board member, a regulator, a competitor.

Steelmanning

Numbers need code

If a number appears and no code produced it, it was predicted. Do this instead.

> **Prompt**
>
> `Compute this in Python. Do not report a number you did not compute with code.`

Asked to sum three numbers 83, 16, 104, AI is much better at predicting the code `sum([83, 16, 104])` than it is at predicting what should follow `83 + 16 + 104 =`

Once code runs correctly, save it. Re-use it rather than rewrite it.

The practice

- Save working scripts as `.py` files
- “Use `my_script.py` — don’t rewrite the code”
- Tested code is known-good; new code is new risk

Later, we will save code in conjunction with skills.

> The principle behind software libraries: write once, test, reuse.

Re-use tested code

Hallucinations

AI predicts text. It does not check what it produces against a source.

Hallucinations are most likely when it is asked to recall something specific that is not in front of it:

- Citations, quotes
- Numbers about a particular company, year, or dataset

Verify the facts

Ask for a source and look it up.

Hallucinations in code

Caught by running it

A made-up function or argument raises an error as soon as the code runs. The agent reads the traceback and fixes it.

Not caught by running it

The wrong column, a join that drops rows, a filter on the wrong year. The code runs and returns a number.

> The second kind usually results from a misunderstanding or an unstated assumption, not a hallucination. Discussing the plan beforehand and the decisions afterwards is what surfaces it.